RIT | جامعة روتشستر الأمريكية للتكنولوجيا في دبي
A Global American University in Dubai

GCIS 124 - 601

Spring 2026

Assignment 2 Comparison and Reflection

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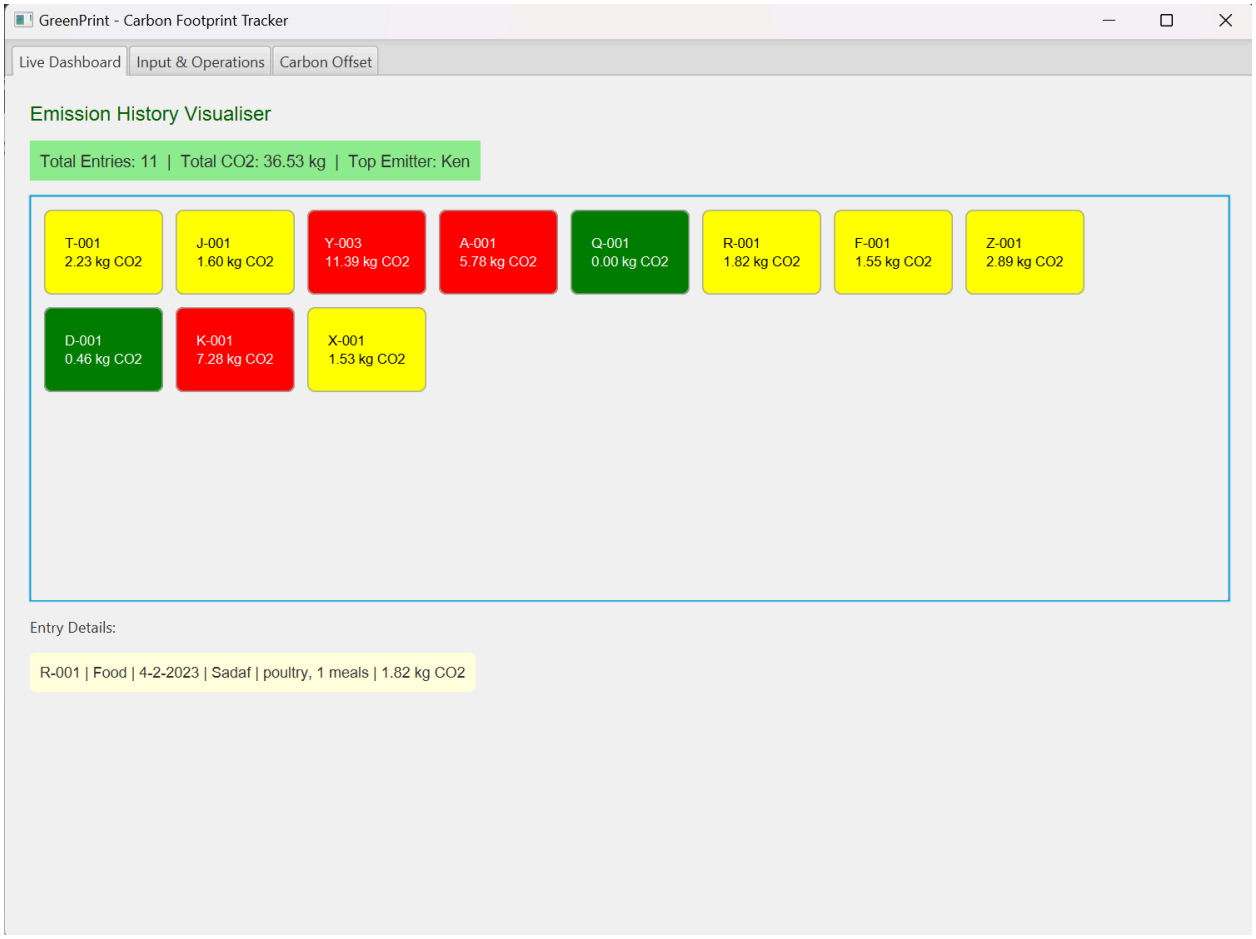
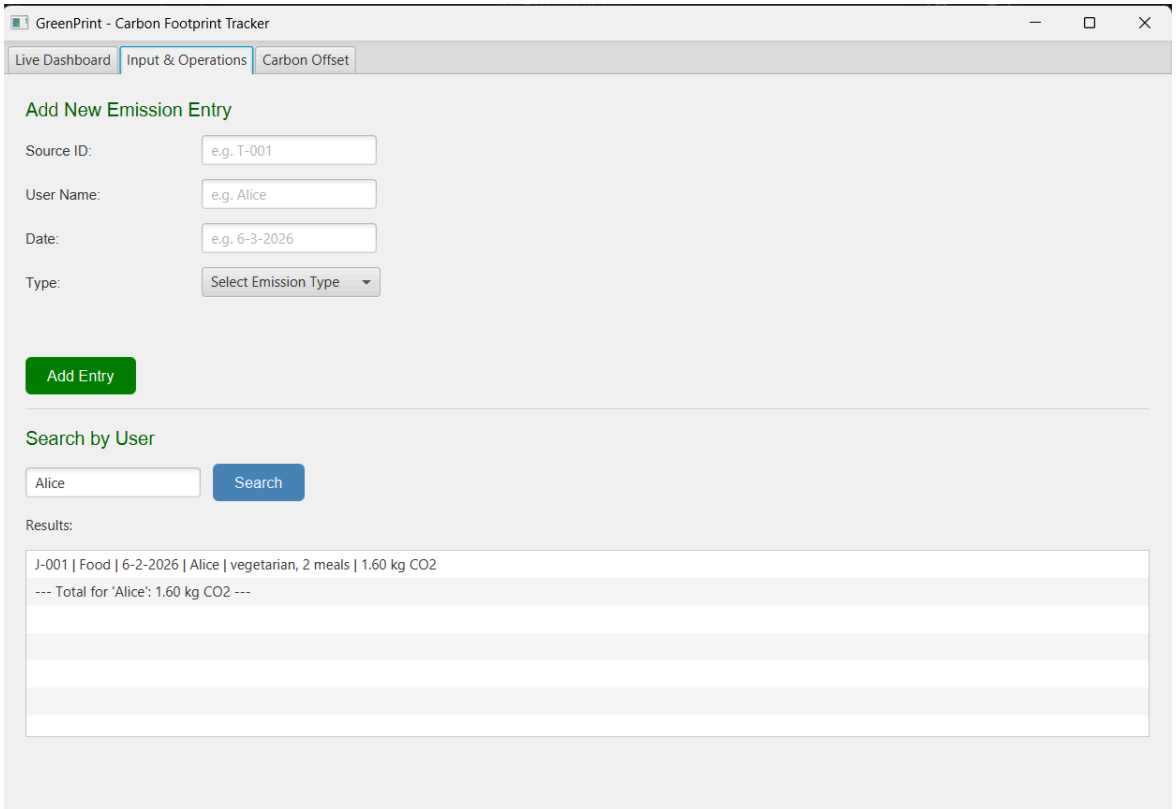
Comparison Table

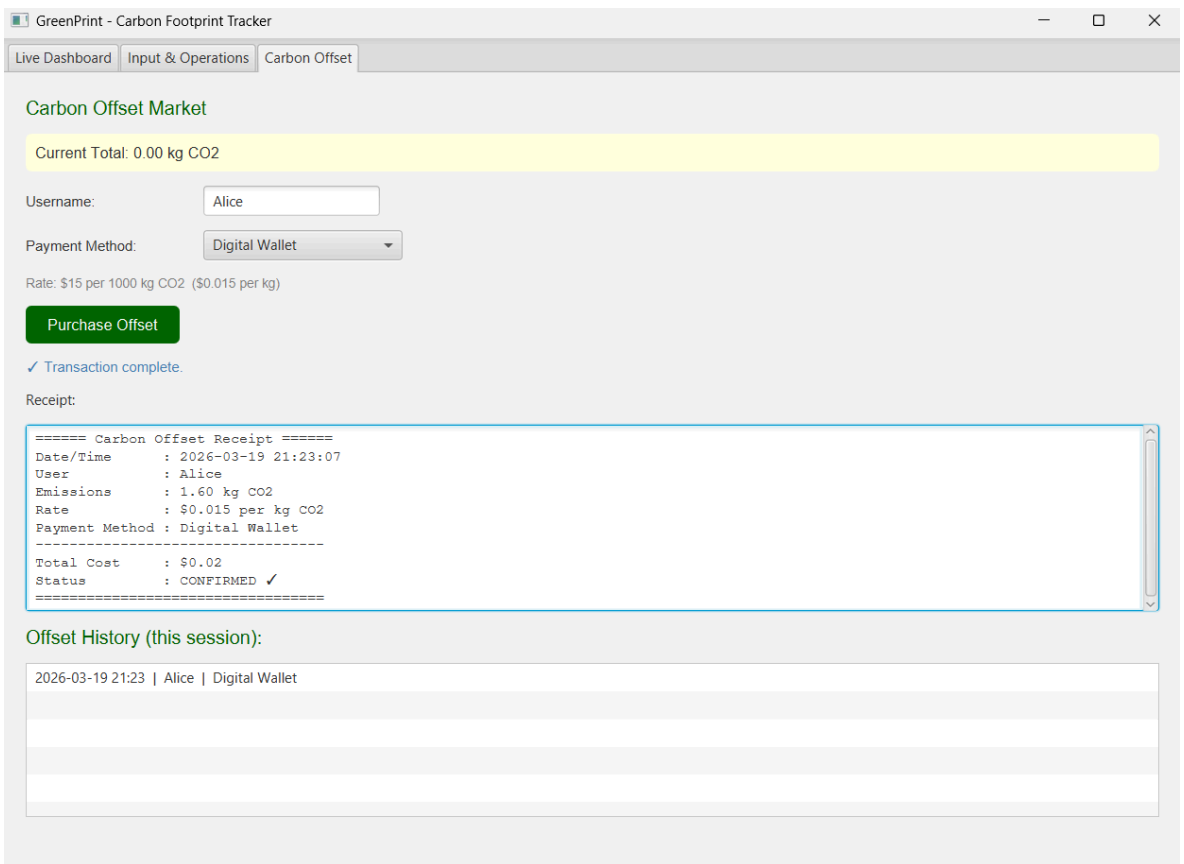
Criterion	Manual GUI (Part A)	AI GUI (Part B)
Code Structure & Readability	The code is well structured and labeled allowing the reader to find blocks of code easily.	Code is more complex, uses CSS, Lambda and switch resulting in a lower number of lines. Has little to no distinct error handling.
GUI Layout & Visual Quality	Dynamic. Uses a FlowPane with custom cards. It's a functional, data-dense visualization of history.	Basic List. Just a standard ListView. It looks "clean" only because it avoided the effort of building a custom grid. Is not visually pleasing, has no sense of aesthetics or taste.
Feature Completeness	Full. Includes the required **Payment Method** ComboBox and session-based history tracking.	Partial. Contains all 3 tabs but does not store session based tracking but rather overall tracking. It is not following the required source ID pattern.
Input Validation & Error Handling	It features Real-time validation with labels (✓/✗) and checks for Unique IDs to prevent data corruption. Uses try-catch, gives immediate feedback and exactly which information.	It reacts only after pressing submit no real time validation. It doesn't specifically try-catch, and update the user of the specific mistake. Instead saying "Recheck all tabs".
Responsiveness & UX	Instant feedback and intuitive navigation between tabs.	Fake. Uses a loading screen to wait, pretending the AI is calculating when it's already done.
Development Time	Real Effort. 3 days of logical mapping, UI alignment, and state management.	Quick but needed guidance. Over 1 hour of prompting to get similar but not fully better results.

AI Prompting Quality	N/A	The prompts themselves were not specific; we used basic language and didn’t include detailed instructions. As a result, it required more iterations and continuous testing and re-prompting to achieve the final result we were satisfied with.
Required Human Edits to AI Output	N/A	Over 10-12 human edits were needed in incorrect naming and changing the formatting to make it easier to understand.

Screenshots of Output GUI (Manual)

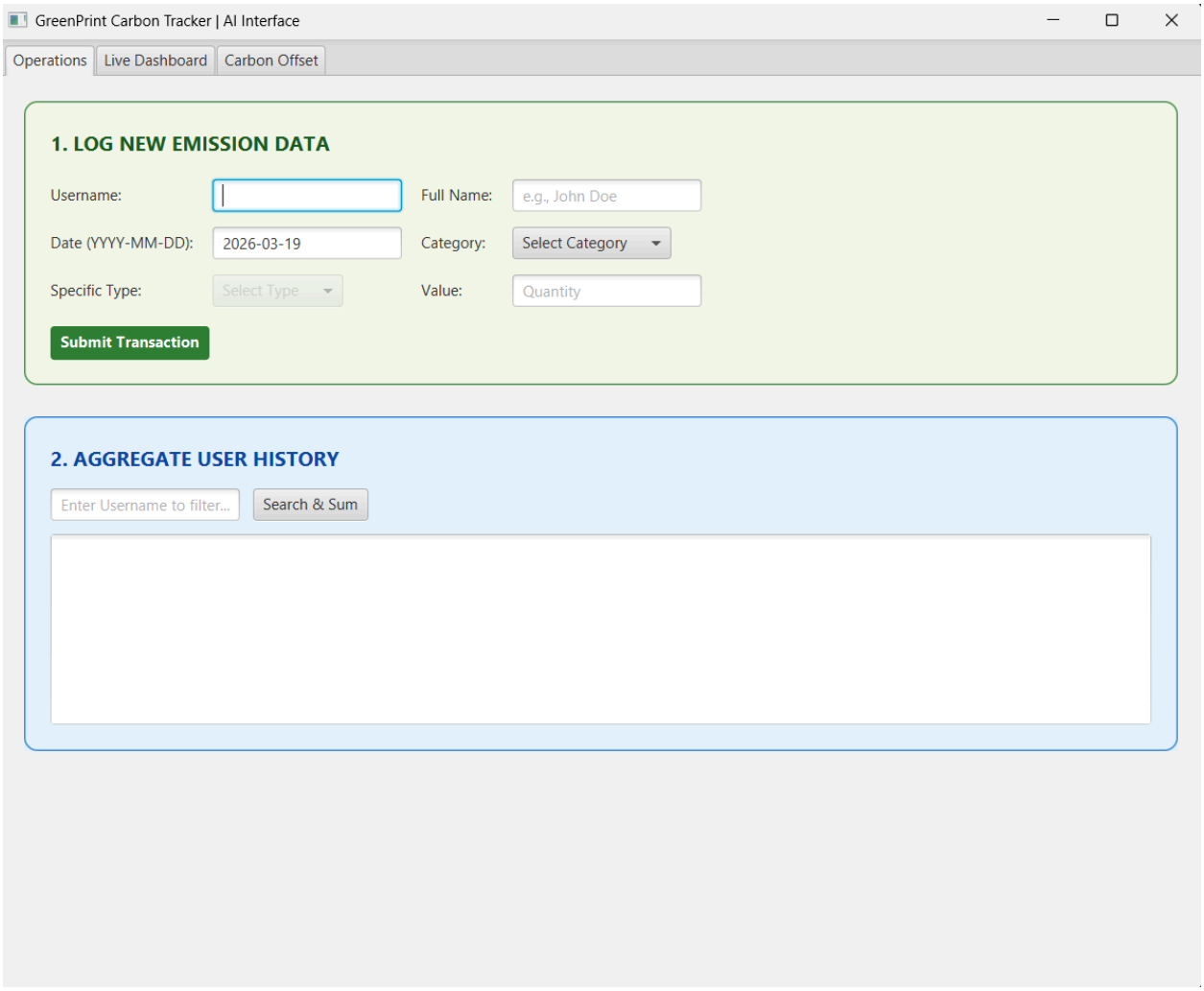


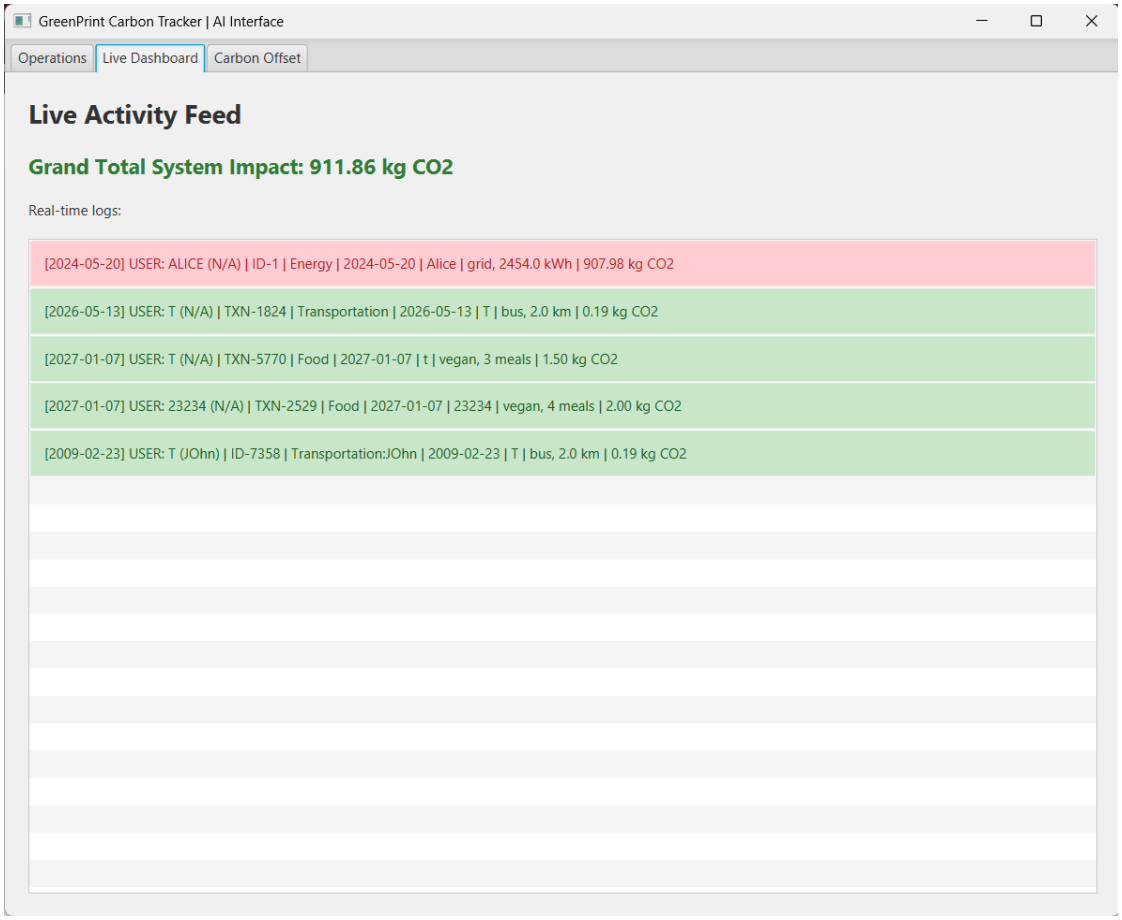

GreenPrint - Carbon Footprint Tracker
Live Dashboard
Input & Operations
Carbon Offset
Add New Emission Entry
Source ID:
e.g. T-001
User Name:
e.g. Alice
Date:
e.g. 6-3-2026
Type:
Select Emission Type
Add Entry
Search by User
Alice
Search
Results:
J-001 | Food | 6-2-2026 | Alice | vegetarian, 2 meals | 1.60 kg CO2
--- Total for 'Alice': 1.60 kg CO2 ---

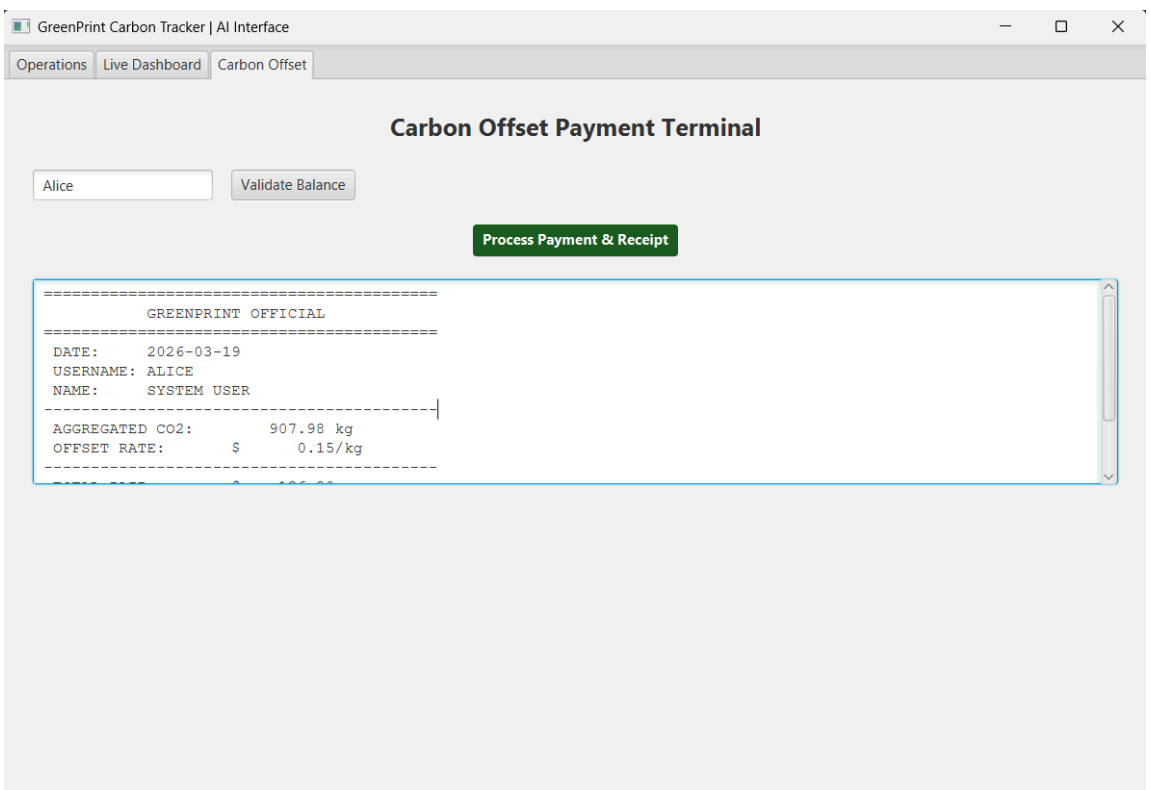

GreenPrint - Carbon Footprint Tracker
Live Dashboard
Input & Operations
Carbon Offset
Carbon Offset Market
Current Total: 0.00 kg CO2
Username:
Alice
Payment Method:
Digital Wallet
Rate: $15 per 1000 kg CO2 ($0.015 per kg)
Purchase Offset
✓ Transaction complete.
Receipt:
====== Carbon Offset Receipt ======
Date/Time : 2026-03-19 21:23:07
User : Alice
Emissions : 1.60 kg CO2
Rate : $0.015 per kg CO2
Payment Method : Digital Wallet

Total Cost : $0.02
Status : CONFIRMED ✓
==========
Offset History (this session):
2026-03-19 21:23 | Alice | Digital Wallet

Screenshots of Output GUI (AI)




GreenPrint Carbon Tracker | AI Interface
Operations
Live Dashboard
Carbon Offset
Live Activity Feed
Grand Total System Impact: 911.86 kg CO2
Real-time logs:
[2024-05-20] USER: ALICE (N/A) | ID-1 | Energy | 2024-05-20 | Alice | grid, 2454.0 kWh | 907.98 kg CO2
[2026-05-13] USER: T (N/A) | TXN-1824 | Transportation | 2026-05-13 | T | bus, 2.0 km | 0.19 kg CO2
[2027-01-07] USER: T (N/A) | TXN-5770 | Food | 2027-01-07 | t | vegan, 3 meals | 1.50 kg CO2
[2027-01-07] USER: 23234 (N/A) | TXN-2529 | Food | 2027-01-07 | 23234 | vegan, 4 meals | 2.00 kg CO2
[2009-02-23] USER: T (JOhn) | ID-7358 | Transportation:JOhn | 2009-02-23 | T | bus, 2.0 km | 0.19 kg CO2


GreenPrint Carbon Tracker | AI Interface
Operations
Live Dashboard
Carbon Offset
Carbon Offset Payment Terminal
Alice
Validate Balance
Process Payment & Receipt
==========
GREENPRINT OFFICIAL
==========
DATE: 2026-03-19
USERNAME: ALICE
NAME: SYSTEM USER
AGGREGATED CO2: 907.98 kg
OFFSET RATE: $ 0.15/kg

Reflection

The AI-generated code did not use multiple try-catch blocks and instead handled errors as a single issue. This made it harder for the user to understand exactly what mistake they were making. In contrast, the manually written code used multiple try-catch blocks, which clearly showed specific errors and provided more useful feedback. It also offered live feedback, while the AI code only returned feedback after execution. A lot of AI code gave out errors only after manual fixing and more prompt generation did we get a reasonable code.

Error handling in AI-generated code was more difficult, as being familiar with our own code made debugging easier. Another issue was that the AI often changed large parts of the code based on new prompts, whereas manual coding allowed us to make small, precise changes. It also provided a good base that could be used as a starting point for further development. It introduced methods we were not familiar with, which helped us learn new concepts.

Additionally, we faced difficulties with parsing, event handlers, semantic errors, and error handling in general. However, the AI was able to produce similar results in much less time, around 10 minutes, compared to about 3 days for the manual code..

We would use AI to help make a scaffolding for our code as it gives us a base and support. However, we would not use it for details as it becomes frustrating to repeatedly prompt the AI for what we need.